Coarsening and Arrest in Colloidal Gels: Modelling via Brownian Dynamics

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Abstract

This project explores the dynamics of colloidal gels, focusing on the processes of coarsening and arrest. We construct a minimal model of attractive Brownian particles and simulate their dynamics using an Euler–Maruyama scheme. The growth of the structure factor $S(q, t)$ is analyzed in the presence and absence of gelation. The theoretical framework draws primarily on the works of Doi [1, 2] for statistical mechanics and colloidal diffusion, and on the works of van Saarloos et al. on ageing and non-equilibrium states.

Contents

1 Introduction

Colloidal suspensions — systems composed of particles with sizes between 1 nm and 1 µm dispersed in a solvent — are a privileged playground for the physics of non-equilibrium condensed matter. When sufficiently strong attractive interactions are induced between these particles (for example by the addition of salt in a charged suspension, by polymer depletion, or by unscreened van der Waals interactions), the system may undergo phase separation towards a dense state. However, the intrinsically slow dynamics of massive colloidal particles can trap the system in a non-equilibrium state *before* phase separation is complete: this is the phenomenon of **gelation as arrested phase separation** [3, 4].

The goal of this work is to construct a minimal model of attractive Brownian particles, to simulate its dynamics using an Euler–Maruyama scheme, and to analyse the growth of the structure factor $S(q,t)$ in the presence and absence of gelation. The theoretical framework draws primarily on the works of Doi [1, 2] for statistical mechanics and colloidal diffusion, and on the works of van Saarloos et al. on ageing and non-equilibrium states.

2 Brownian Motion and Colloidal Diffusion

2.1 The Langevin Equation

Following Doi (Ch. 6), the motion of a colloidal particle suspended in a solvent is described by separating the force exerted by the fluid into two contributions:

- a deterministic viscous friction force,
- a stochastic random force arising from microscopic collisions with solvent molecules.

The equation of motion is the Langevin equation:

$$m\ddot{\boldsymbol{r}} = -\zeta\dot{\boldsymbol{r}} - \nabla U + \boldsymbol{F}_r(t), \tag{1}$$

where:

- $-\zeta\dot{\boldsymbol{r}}$ is the viscous friction force,
- $U(\boldsymbol{r})$ is the interaction potential,
- $\boldsymbol{F}_r(t)$ is the stochastic random force generated by thermal fluctuations of the solvent.

For a spherical colloid of radius a suspended in a solvent of viscosity η, the Stokes friction coefficient is:

$$\zeta = 6\pi\eta a. \tag{2}$$

The random force satisfies:

$$\langle \boldsymbol{F}_r(t)\rangle = 0, \tag{3}$$

meaning that thermal fluctuations have zero average.

Its temporal correlations are:

$$\langle F_{r,\mu}(t)F_{r,\nu}(t')\rangle = 2\zeta k_B T\delta_{\mu\nu}\delta(t-t'), \tag{4}$$

where $\mu,\nu \in \{x,y,z\}$ denote Cartesian components.

Equation (4) expresses the **fluctuation–dissipation theorem**: the same solvent responsible for viscous dissipation also generates thermal fluctuations.

The Dirac delta function indicates that the random force is **delta-correlated in time**: collisions occurring at different times rapidly become decorrelated because the microscopic configuration of the solvent changes on extremely short timescales.

2.2 Overdamped Limit: Brownian Dynamics

For colloidal particles suspended in a liquid, viscous dissipation dominates inertia.

The inertial relaxation time:

$$\tau_m = \frac{m}{\zeta} \tag{5}$$

is typically much smaller than the diffusive timescale:

$$\tau_D = \frac{a^2}{D_0}. \tag{6}$$

For a colloid of radius 100 nm in water:

$$\tau_m \sim 10^{-8}\ \text{s}, \qquad \tau_D \sim 10^{-3}\ \text{s}. \tag{7}$$

Therefore:

$$\tau_m \ll \tau_D, \tag{8}$$

meaning that the particle velocity relaxes almost instantaneously.

The inertial term in Eq. (1) can thus be neglected:

$$m\ddot{\boldsymbol{r}} \simeq 0. \tag{9}$$

The Langevin equation reduces to:

$$\boxed{\zeta\dot{\boldsymbol{r}} = -\nabla U + \boldsymbol{F}_r(t)} \tag{10}$$

This equation constitutes the basis of **Brownian dynamics**.

Using the Einstein relation:

$$D_0 = \frac{k_B T}{\zeta}, \tag{11}$$

Eq. (10) becomes:

$$\dot{\boldsymbol{r}} = -\frac{D_0}{k_B T}\nabla U + \frac{1}{\zeta}\boldsymbol{F}_r(t). \tag{12}$$

This form clearly separates:

- deterministic drift induced by interactions,
- stochastic diffusion induced by thermal agitation.

The solvent viscosity enters explicitly through the friction coefficient ζ and controls the particle mobility:

$$\mu = \frac{1}{\zeta}. \tag{13}$$

Increasing viscosity decreases diffusion and slows down colloidal dynamics.

3 Colloidal Interactions

3.1 Pair Potentials between Colloids

The physics of colloidal systems rests on the competition between several pair interactions. Following the framework of Doi (Ch. 2), the interaction between two particles is described by a pair potential $U(r)$, where r is the centre-to-centre distance.

3.1.1 Hard-Core Repulsion

Two particles of radius a cannot overlap. This is represented by a hard-core potential:

$$U_{\mathrm{HS}}(r) = \begin{cases} +\infty & r < 2a \\ 0 & r \geq 2a \end{cases} \tag{14}$$

In practice, simulations often replace this by a soft polynomial or exponential repulsion to guarantee continuity of forces.

3.1.2 Van der Waals Interactions

Quantum fluctuations of electronic dipoles induce a long-range attractive interaction between dielectric materials (London–van der Waals interaction). For two spheres of radius a and Hamaker constant A_H, the van der Waals potential is [5]:

$$U_{\mathrm{vdW}}(r) = -\frac{A_H}{6}\left[\frac{2a^2}{r^2 - 4a^2} + \frac{2a^2}{r^2} + \ln\left(1 - \frac{4a^2}{r^2}\right)\right]. \tag{15}$$

This potential diverges as $r \to 2a$, making electrostatic stabilisation of colloids essential.

3.1.3 Electrostatic Interaction: DLVO Theory

For charged particles in an electrolyte, electrostatic repulsion is described by the Derjaguin–Landau–Verwey–Overbeek (DLVO) theory. Each particle is surrounded by an electrical double layer of Debye length κ^{-1}, defined by:

$$\kappa^2 = \frac{e^2}{\varepsilon_0 \varepsilon_r k_B T} \sum_i n_i^0 z_i^2, \tag{16}$$

where n_i^0 is the bulk concentration of ionic species i of valence z_i, ε_r the relative permittivity of the solvent, and $k_B T$ the thermal energy. The repulsive potential between two spheres of surface potential ψ_0 reads, in the Debye–Hückel limit:

$$U_{\mathrm{el}}(r) = 4\pi\varepsilon_0\varepsilon_r a^2 \psi_0^2 \, \frac{e^{-\kappa(r-2a)}}{r}. \tag{17}$$

The total DLVO potential is $U_{\mathrm{DLVO}} = U_{\mathrm{vdW}} + U_{\mathrm{el}}$. Adding salt increases κ, reduces the Debye length, lowers the electrostatic barrier, and can render interactions *effectively* attractive at short range: this is the mechanism of salt-induced gelation.

3.2 Choice of Potential for Simulations

For our minimal model we adopt the **Morse potential**:

$$\boxed{U_{\mathrm{M}}(r) = \epsilon\left[e^{-2\alpha(r-r_0)} - 2\,e^{-\alpha(r-r_0)}\right],} \tag{18}$$

where $\epsilon > 0$ is the well depth, $r_0 = 2a$ is the equilibrium (contact) distance, and α controls the range of attraction. The derived force is:

$$\boldsymbol{F}_{\mathrm{M}}(\boldsymbol{r}) = -\nabla U_{\mathrm{M}}(r) = \frac{2\alpha\epsilon}{r}\left[e^{-2\alpha(r-r_0)} - e^{-\alpha(r-r_0)}\right]\boldsymbol{r}. \tag{19}$$

This choice has several advantages:

- The potential is smooth (differentiable everywhere), unlike the hard-core or truncated Lennard-Jones potentials.
- The parameter α allows independent control of the range and well depth, enabling realistic modelling of short-range colloidal interactions.
- The force has a simple analytic form, which is favourable for numerical integration.
- For $\alpha(r - r_0) \gg 1$, the potential becomes negligible, facilitating truncation at a cutoff distance r_c.

In practice, we choose $r_c = r_0 + 5/\alpha$ (approximately five times the characteristic interaction range), and apply a shift $U \to U - U(r_c)$ to ensure continuity.

Remark 1. A widely used alternative is the truncated and shifted Lennard-Jones (WCA) potential supplemented by an attractive well ($r_{\min} < r < r_c$):

$$U_{\mathrm{LJ}}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right], \tag{20}$$

commonly employed in simulations of model colloids [6]. The Morse potential is however preferable here as it avoids the r^{-12} singularity, which forces very small time steps.

4 Phase Separation in Colloidal Suspensions

4.1 Phase Diagram and Spinodal Instability

The phase diagram of a system of attractive hard spheres (an effective van der Waals fluid model) exhibits a liquid–gas-like coexistence, with a critical temperature T_c and critical density ϕ_c. In the (ϕ, T) space where ϕ is the volume fraction, the spinodal curve delimits the region of thermodynamic instability where:

$$\left(\frac{\partial^2 f}{\partial \phi^2}\right)_T < 0, \tag{21}$$

with $f(\phi, T)$ the free energy density. Inside the spinodal, the system is *unconditionally unstable*: any arbitrarily small density fluctuation grows spontaneously, driving **spinodal decomposition**.

4.2 Spinodal Decomposition: the Cahn–Hilliard Theory

Within the Cahn–Hilliard framework, the functional free energy of the system is:

$$\mathcal{F}[\phi] = \int \left[f(\phi) + \frac{\kappa}{2}|\nabla\phi|^2 \right] dV, \tag{22}$$

where the gradient term penalises interfaces. The conservative dynamics of $\phi(\boldsymbol{r}, t)$ (particle number is conserved) is governed by:

$$\frac{\partial\phi}{\partial t} = \nabla \cdot \left[M(\phi)\, \nabla \frac{\delta\mathcal{F}}{\delta\phi} \right] = \nabla \cdot \left[M(\phi)\, \nabla\big(f'(\phi) - \kappa\nabla^2\phi\big) \right], \tag{23}$$

where $M(\phi) = D_0\phi(1-\phi)/(k_BT)$ is the mobility. Linearising around a uniform composition ϕ_0:

$$\frac{\partial\delta\phi}{\partial t} = -M_0\kappa\, q^4\, \delta\phi + M_0 f''(\phi_0)\, q^2\, \delta\phi, \tag{24}$$

one obtains the growth rate for a mode of wavevector q:

$$\sigma(q) = M_0 q^2 \left[f''(\phi_0) - \kappa q^2 \right]. \tag{25}$$

For $f''(\phi_0) < 0$ (spinodal region), $\sigma(q) > 0$ for modes $q < q_c = \sqrt{|f''(\phi_0)|/\kappa}$, meaning that long-wavelength fluctuations (small q) grow exponentially. The most unstable wavelength is $q^* = q_c/\sqrt{2}$, corresponding to the characteristic initial length scale of the decomposition:

$$\ell^* = \frac{2\pi}{q^*} = 2\pi\sqrt{\frac{2\kappa}{|f''(\phi_0)|}}. \tag{26}$$

5 Gelation as Arrested Phase Separation

5.1 General Mechanism

Gelation is often described as a *percolation transition* in a network of interparticle bonds. However, the experimental work of Trappe et al. [3] and the theoretical perspective introduced by Poon, Pusey and Lekkerkerker, then formalised by Lu et al. [4], showed that colloidal gelation is **a spinodal phase separation whose kinetics is arrested** by the dynamic arrest of the dense phase.

The scenario proceeds as follows:

1. The system is quenched into the spinodal region (by increasing the ionic strength, or by increasing the depth of the attractive well).
2. Spinodal decomposition begins: particle-rich domains grow.
3. When the local volume fraction in the dense phase reaches the glass transition volume fraction ϕ_g, particles can no longer rearrange: the dense phase **vitrifies** (dynamic arrest).
4. Domain growth is **arrested**: the percolating network of dynamically frozen particles constitutes the *colloidal gel*.

This mechanism is distinct from gelation by simple bond percolation (as in a polymer network). It is fundamentally out of equilibrium and depends on the *rate* of the quench as well as the depth of the spinodal region.

5.2 Arrest Length and Arrest Criterion

The arrest length ℓ_{arrest} is determined by the competition between the domain growth time τ_{grow} and the structural relaxation time of the dense phase τ_{relax}. When $\tau_{\text{relax}} \to \infty$ (dynamic arrest), the characteristic domain length freezes at ℓ_{arrest}. The structure factor $S(q,t)$ then shows a stable peak at:

$$q^* \sim \ell_{\text{arrest}}^{-1}, \tag{27}$$

whose position and amplitude no longer grow with time — the experimental and numerical signature of arrest.

5.3 Ageing and Non-Equilibrium States

From van Saarloos's perspective, colloidal gels are **ageing non-equilibrium states**: their properties (rheology, diffusion, structure) depend explicitly on the ageing time t_w elapsed since preparation. The mean-squared displacement of a particle in the gel exhibits an intermediate **plateau** characteristic of cage trapping (glassy behaviour), possibly followed by long-time relaxation (cage hopping):

$$\langle |\Delta \boldsymbol{r}(t)|^2 \rangle \xrightarrow{t\to\infty} \text{plateau} \sim r_{\text{cage}}^2 \qquad \text{(gel)}. \tag{28}$$

The slow dynamics is often described by a KWW (Kohlrausch–Williams–Watts) relaxation:

$$C(t) = \exp\left[-(t/\tau_\alpha)^\beta\right], \quad 0 < \beta < 1, \tag{29}$$

where τ_α is the α-relaxation time and $\beta < 1$ indicates stretched (non-exponential) relaxation.

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6 Structure Factor, Coarsening and Dynamic Scaling

6.1 Time-Dependent Structure Factor

The central observable of this study is the time-dependent structure factor $S(q,t)$, which characterises the spatial organisation of the colloidal suspension during phase separation and gelation.

For a system of N particles at positions $\boldsymbol{r}_j(t)$, the instantaneous density mode is:

$$\rho(\boldsymbol{q},t) = \sum_{j=1}^{N} e^{i\boldsymbol{q}\cdot\boldsymbol{r}_j(t)}. \tag{30}$$

The structure factor is then defined as:

$$S(\boldsymbol{q},t) = \frac{1}{N}\left|\rho(\boldsymbol{q},t)\right|^2 = \frac{1}{N}\left|\sum_{j=1}^{N} e^{i\boldsymbol{q}\cdot\boldsymbol{r}_j(t)}\right|^2. \tag{31}$$

After spherical averaging, one obtains the isotropic quantity $S(q,t)$.

The structure factor is directly related to density correlations in the system. At early times, when the suspension remains approximately homogeneous, $S(q,t)$ is relatively flat.

As phase separation develops, a peak progressively emerges at a characteristic wavevector $q^*(t)$ corresponding to the dominant domain size.

The characteristic domain length is therefore defined by:

$$\boxed{\ell(t) = \frac{2\pi}{q^*(t)}.} \tag{32}$$

The growth of $\ell(t)$ quantifies the coarsening of colloidal domains.

6.2 Diffusive Coarsening

Following the initial spinodal decomposition, domains rich and poor in colloids progressively coarsen in order to reduce interfacial free energy. For diffusion-controlled phase separation, Lifshitz and Slyozov showed that the characteristic domain size obeys the asymptotic growth law:

$$\ell(t) \propto t^{1/3}, \tag{33}$$

known as the Lifshitz–Slyozov law.

This scaling law reflects the fact that domain growth is limited by diffusive transport between interfaces. The exponent $1/3$ is universal for diffusion-limited coarsening, whereas the prefactor depends on microscopic properties such as diffusion and interfacial tension.

As coarsening proceeds:

- the diffraction peak of $S(q,t)$ shifts towards smaller values of q,
- the characteristic length $\ell(t)$ increases,
- density correlations become stronger and more spatially extended.

6.3 Dynamic Scaling

At sufficiently long times, the system enters a dynamic scaling regime in which the morphology becomes statistically self-similar. The structure factor then obeys the scaling form:

$$S(q,t) = \ell(t)^d \, \widetilde{S}(q\ell(t)), \tag{34}$$

where d is the spatial dimension and $\widetilde{S}$ is a universal scaling function.

Equation (34) implies that all structure factors measured at different times collapse onto a single master curve when plotted as:

$$\frac{S(q,t)}{\ell(t)^d} \quad \text{versus} \quad q\ell(t). \tag{35}$$

This collapse is the hallmark of self-similar coarsening dynamics.

6.4 Arrested Phase Separation and Gelation

In ordinary spinodal decomposition, coarsening proceeds continuously and $\ell(t)$ keeps increasing according to Eq. (33).

In contrast, colloidal gelation corresponds to an arrested phase separation process. When the dense colloid-rich domains become dynamically arrested, particle rearrangements slow dramatically and domain growth stops.

As a consequence:

$$\ell(t) \to \ell_{\text{arrest}}, \tag{36}$$

where ℓ_{arrest} is the arrest length characterising the gel network.

Equivalently, the diffraction peak of $S(q,t)$ stabilises at a finite wavevector:

$$q^* \sim \ell_{\text{arrest}}^{-1}, \tag{37}$$

whose position and amplitude no longer evolve significantly with time.

The comparison between free coarsening and arrested coarsening therefore constitutes the main observable of the present work.

7 Numerical Implementation

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7.1 Euler–Maruyama Scheme

The overdamped Langevin equation eq. (12) is integrated numerically using the Euler–Maruyama scheme, which is the standard discretisation method for stochastic differential equations with additive noise [7].

Integrating the Langevin equation over a finite time step Δt gives two distinct contributions to the particle displacement:

- a deterministic drift induced by interaction forces,
- a stochastic displacement generated by thermal fluctuations.

The deterministic contribution is proportional to Δt, whereas the stochastic contribution results from the accumulation of many independent microscopic collisions with solvent molecules. According to the central limit theorem, this random displacement is Gaussian, with a variance proportional to Δt. Its typical amplitude therefore scales as $\sqrt{\Delta t}$.

For a time step Δt, the Euler–Maruyama update rule becomes:

$$\boxed{\boldsymbol{r}_i(t+\Delta t) = \boldsymbol{r}_i(t) + \frac{D_0}{k_B T}\boldsymbol{F}_i\big(\{\boldsymbol{r}_j(t)\}\big)\Delta t + \sqrt{2D_0\Delta t}\,\boldsymbol{W}_i,} \tag{38}$$

where $\boldsymbol{W}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ is a Gaussian random vector of zero mean and unit variance, independently drawn for each particle and each time step.

The first term corresponds to deterministic drift towards lower-energy configurations, while the second term represents Brownian diffusion due to thermal agitation of the solvent.

This numerical scheme preserves the diffusive behaviour:

$$\langle |\Delta \boldsymbol{r}|^2 \rangle = 2dD_0\Delta t, \tag{39}$$

which is the defining property of Brownian motion.

7.2 Time-Step Selection

The time step Δt must satisfy several criteria:

- **Thermal stability**: $D_0 \Delta t/\sigma^2 \ll 1$, where $\sigma = 2a$ is the particle diameter.
- **Force resolution**: $|F_{\max}|\,\Delta t/\gamma \ll a$, to prevent a particle from crossing another's core in a single step.
- **Temporal resolution**: $\Delta t \ll \tau_{\text{relax}}$, the structural relaxation time of interest.

In practice:

$$\Delta t^* = \min\left(\frac{0.01\,\sigma^2}{D_0},\ \frac{0.01\,\sigma}{|F_{\max}|/\gamma}\right). \tag{40}$$

7.3 Periodic Boundary Conditions

We simulate N particles in a cubic box of side L with **periodic boundary conditions** (PBC). The distance between particles i and j is computed using the minimum image convention:

$$\Delta \boldsymbol{r}_{ij} = \boldsymbol{r}_{ij} - L\,\text{round}\left(\frac{\boldsymbol{r}_{ij}}{L}\right), \tag{41}$$

and the potential cutoff $r_c \leq L/2$ is required to prevent a particle from interacting with its own image.

7.4 Reduced Units

To simplify the simulations, we work in reduced units:

$$r^* = r/\sigma, \quad \sigma = 2a, \tag{42}$$

$$t^* = t\,D_0/\sigma^2, \tag{43}$$

$$U^* = U/k_B T, \tag{44}$$

$$F^* = F\,\sigma/k_B T. \tag{45}$$

In these units, eq. (38) becomes:

$$\boldsymbol{r}_i^*(t^* + \Delta t^*) = \boldsymbol{r}_i^*(t^*) + \boldsymbol{F}_i^*\big(\{\boldsymbol{r}_j^*\}\big)\,\Delta t^* + \sqrt{2\Delta t^*}\,\boldsymbol{W}_i, \tag{46}$$

an equation free of explicit dimensional parameters.

7.5 Simulation Protocol

1. **Initialisation**: positions $\boldsymbol{r}_i^*(0)$ drawn randomly (or on a lattice) in the box, with an overlap check.
2. **Equilibration**: simulation at $\epsilon \to 0$ (hard spheres) or $\epsilon < \epsilon_c$ (outside the spinodal region) to reach an equilibrated liquid reference state.
3. **Quench**: instantaneous increase of ϵ at $t = 0$ to enter the spinodal region.
4. **Production**: integration of eq. (38) for a total time $T_{\text{sim}} \gg \tau_D$.
5. **Analysis**: computation at regular intervals of $S(q,t)$, $g(r,t)$, and $\langle|\Delta \boldsymbol{r}|^2\rangle(t)$.

To study the effect of gelation, two sets of simulations are performed:

- **Free system**: moderate ϵ/k_BT (below the gelation threshold); phase separation proceeds without arrest.
- **Gelled system**: large ϵ/k_BT; the dense phase reaches ϕ_g and domain growth is arrested.

8 Discussion: Comparison of Observables

8.1 Numerical Gelation Criterion

Gelation is detected numerically by several indicators:

1. **Structure factor plateau**: $\partial_t S(q^*, t) \approx 0$.
2. **Mean-squared displacement plateau**: $\langle|\Delta \boldsymbol{r}|^2\rangle(t)$ saturates at a finite value (cf. eq. (28)).
3. **Percolation**: the bond network (particles i and j are bonded if $r_{ij} < r_{\text{bond}}$ and $U(r_{ij}) < 0$) spans from one side of the box to the other.

8.2 Dynamic Scaling and Its Breakdown

For the *free* system, the scaling law eq. (34) is verified by superimposing the curves $S(q,t)/\ell(t)^d$ versus $q\ell(t)$. For the *gelled* system, this law breaks down: the curves do not collapse, which is the signature of the scale-invariance violation caused by arrest.

9 Conclusion

We have presented the complete theoretical framework for studying colloidal gelation as arrested phase separation. The central point is that Brownian dynamics, described by the overdamped Langevin equation and integrated by the Euler–Maruyama scheme, faithfully reproduces the physics of colloidal diffusion and aggregation. The competition between spinodal dynamics (domain growth as $t^{1/3}$) and kinetic arrest (vitrification of the dense phase) gives rise to the colloidal gel, observable through the arrest of the structure factor peak. The Morse potential, by virtue of its smoothness and tuneable range, constitutes a minimal yet physically realistic choice for capturing these effects.

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